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Paper Code : PEC-IT602B Data Warehousing and Data Mining

UPID : 006584

Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.*

*Candidate are required to give their answers in their own words as far as practicable*

## Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following :

[ 1 x 10 = 10 ]

- (I) Data warehouse contains ..... data that is never found in the operational environment.
  - a) normalized    b) informational    c) summary    d) denormalized
- (II) In the measurement of the secular trend, the moving averages:
  - a) Smooth out the time series
  - b) Give the trend in a straight line
  - c) Measure the seasonal variations
  - d) None of these
- (III) The following given statement can be considered as the examples of \_\_\_\_\_.  
Suppose one wants to predict the number of newborns according to the size of storks' population by performing supervised learning.
- (IV) Text mining is being used by large media companies, to clarify information and to provide readers with greater search experiences,
  - a) TRUE
  - b) FALSE
- (V) Which of the following is the correct advantage of the Update-Driven Approach?
  - a) This approach provides high performance.
  - b) The data can be copied, processed, integrated, annotated, summarized and restructured in the semantic data store in advance.
  - c) Both 1 and 2
  - d) None of the above
- (VI) Removing duplicate records is a process called .....
  - a) recovery
  - b) data pruning
  - c) data cleaning
  - d) data cleansing
- (VII) Effect of one attribute value on a given class is independent of values of other attribute is called \_\_\_\_\_.
  - a) value independence
  - b) class conditional independence
  - c) conditional independence
  - d) unconditional independence
- (VIII) A set of observations recorded at an equal interval of time is called
  - a) Array data
  - b) Data
  - c) Geometric Series
  - d) Time series data
- (IX) For what purpose, the analysis tools pre-compute the summaries of the huge amount of data?
- (X) What are the functions of Data Mining?
- (XI) Which of the following can be considered as the drawback of the query-Driven approach in data warehousing?
  - a) This approach is expensive for queries that require aggregations
  - b) This approach is expensive insufficient, and very frequent queries
  - c) This approach requires a very complex integration and filtering process
  - d) All of the above
- (XII) The type of relationship in star schema is .....
  - a) many to many

- b) one to one
- c) one to many
- d) many to one

### Group-B (Short Answer Type Question)

Answer *any three* of the following :

[ 5 x 3 = 15 ]

2. What is association rule learning algorithm? Define support and confidence. [5]
3. Compare and contrast the data warehouse and operational DB with various features. [5]
4. What is time series analysis? [5]
5. What is spatial mining? [5]
6. Explain Class Imbalance Problem. [5]

### Group-C (Long Answer Type Question)

Answer *any three* of the following :

[ 15 x 3 = 45 ]

7. (a) Differentiate classification and prediction. [ 6 ]  
 (b) Mention the advantages of hierarchical clustering. Define dendrogram [ 6 ]  
 (c) What is association Rule mining? Explain with example. [ 3 ]
8. (a) What are the major challenges faced in bringing data mining research to market? [ 5 ]  
 (b) What are Research prototypes? [ 5 ]  
 (c) Explain graph mining with an example. [ 5 ]
9. (a) You are given the transaction data shown in the Table below from a fast food restaurant. There are 9 distinct transactions (order: 1–order: 9) and each transaction involves between 2 and 4 meal items.  
 There are a total of 5 meal items that are involved in the transactions. For simplicity we assign the meal items short names (M1–M5) rather than the full descriptive names (e.g., Big Mac).  
 Apply the Apriori algorithm for discovering frequent item sets of the following. Use minimum support value is 3.  
 order meal items  
 Order: 1 M1, M2, M5  
 Order: 2 M2, M4  
 Order: 3 M2, M3  
 Order: 4 M1, M2, M4  
 Order: 5 M1, M3  
 Order: 6 M2, M3  
 Order: 7 M1, M3  
 Order: 8 M1, M2, M3, M5  
 Order: 9 M1, M2, M3  
 (b) What is data reduction? [ 3 ]  
 (c) Briefly discuss the schemas for multidimensional databases. [ 4 ]
10. (a) Use single link agglomerative clustering to group the data described by the following distance matrix. Show the dendrograms. <https://www.makaut.com> [ 8 ]
 

|   | A | B | C | D |
|---|---|---|---|---|
| A | 0 | 1 | 4 | 5 |
| B |   | 0 | 2 | 6 |
| C |   |   | 0 | 3 |
| D |   |   |   | 0 |

  
 (b) Write K-Means algorithm and mention its applications in data mining. [ 7 ]
11. (a) Discuss in detail about data mining applications. [ 5 ]  
 (b) What are outliers analysis in data mining? Discuss the methods adopted for outlier detection. [ 7 ]  
 (c) What are the drawbacks of Apriori Algorithm? [ 3 ]

\*\*\* END OF PAPER \*\*\*